



Production Information Systems

Topic 2 : DataBase Modeling and Design

CHAPTER 3 Data Modeling

ISE Department
Year 4/ Level 7

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Outline

1. Introduction
2. Entity-Relationship (E-R) Modeling
- 3. Normalization**
- 4. Case Study in Data Modeling**

3.Normalization

- When designing a data model, attribute field redundancy must be avoided as much as possible. This is so that the implementation in tables is not burdened with the storage of unnecessary data.
- ***Normalization*** is the name given to the process of evaluating a model in order to control data redundancies.
- Data redundancy causes problems with the integrity of data when the usual database operations of insert, delete and update are performed on the data.
- These problems are called ***anomalies***, most commonly referred to as ***insertion anomalies, deletion anomalies, and update anomalies***.

3.Normalization

Case 1

Entity Set: VENDOR

VENDOR ID	V NAME	V STREET	V CITY	V STATE	V ZIP
V110	Jersey Vegetable Co.	2 Main St.	Patterson	NJ	07055
V25	General Provisions	125 Common St.	Boise	ID	44830
V250	Spices Unlimited	25 Saltv Lane	East Hampton	NY	10027
V75	Pasta Supply, Inc.	34 Henry St.	Philadelphia	PA	09098

Entity Set: PURCHASE_ORDER

PO NUMBER	RELEASE DATE	PO STATUS	PO AMT	VENDOR ID
2591	2/10/06	CLOSED	\$4,300.00	V110
2592	2/10/06	OPEN	\$505.50	V25
2593	2/11/06	OPEN	\$4,000.00	V110
2594	2/12/06	OPEN	\$3,280.00	V250
2595	2/15/06	OPEN	\$500.00	V250
2596		HOLD	\$1,000.00	V75

Table: PO_Vendor

Case 2

PO NUM	RELEASE DAT	PO STAT	PO AMT	V NAME	V STREET	V CITY	V STATE	V ZIP
2591	2/10/2006	CLOSED	\$4,300.00	Jersey Vegetable Co.	2 Main St.	Patterson	NJ	07055
2592	2/10/2006	OPEN	\$505.50	General Provisions	125 Common St.	Boise	ID	44830
2593	2/11/2006	OPEN	\$4,000.00	Jersey Vegetable Co.	2 Main St.	Patterson	NJ	07055
2594	2/12/2006	OPEN	\$3,280.00	Spices Unlimited	25 Saltv Lane	East Hampton	NY	10027
2595	2/15/2006	OPEN	\$500.00	Spices Unlimited	25 Saltv Lane	East Hampton	NY	10027
2596		HOLD	\$1,000.00	Pasta Supply, Inc.	34 Henry St.	Philadelphia	PA	09098

3.Normalization

3.1 Insertion anomalies

Different kinds of anomalies can occur when inserting new records into the **table PO-Vendor** :

For example, if the enterprise qualifies a new vendor and wishes to record vendor information in the database, the new record in PO_Vendor will start from V_Name.

The first five fields will be NULL values until a purchase order is assigned to the vendor. In fact, it will not be possible to enter the vendor record because PO_Number is the primary key attribute and a primary key cannot be NULL.

Therefore, it will not be possible to retain vendor information in PO_Vendor without first assigning a purchase order.

3.Normalization

3.2 Deletion anomalies

The deletion anomaly results in the complete loss of information when a related record is deleted from the database.

For example : If the purchase order number 2596 is canceled and the purchasing agent wishes to delete it from the database. The vendor for this purchase order is Pasta Supply Inc. In fact, this is the only purchase order to which that vendor is assigned.

If the record for PO number 2596 is deleted, all information in the database concerning Pasta Supply Inc will be lost.

3.Normalization

3.3 Update anomalies

- Consider the situation in which a vendor changes its address.
- For the first case, there is one update to the set of attribute values required.
- For the second PO_Vendor table structure, multiple records will need to be changed depending on the number of purchase orders assigned to the vendor.
- In fact, in large databases this may become very unwieldy and introduces a risk of missing values that have to be updated or of introducing different values for the same data element due to data entry errors.

3.Normalization

3.4 Normal Forms

- **Normalization** is the process of controlling data redundancy and eliminating the possibility of the anomalies previously described.
- The process of normalization tries to uncover ***functional dependencies*** between attributes that are being considered for inclusion in a relation and provides the rules to be applied.

3.Normalization

3.4 Normal Forms

- **A functional dependency** between two attributes (A and B) exists if a value of A uniquely determines a value of B.
- Functional dependency can be identified from a data set by identifying a value of A, “a,” and, for each instance of that value, confirming that it is associated with the same value of B, “b.”

(Case 2 Slide 4) : It is clear that for V_Name attribute with a value “Jersey Vegetable Co.,” the corresponding value of the attribute V_Street is “2 Main St.”

*We say that V_Name **functionally** determines V_Street.*

*It can also be said that V_Name is **functionally dependent** on V_Street since each unique value of V_Street determines a unique value of V_Name.*

*At this point it is clear that the tuple (V_Name, V_Street) **belongs in the same relation.***

3.Normalization

3.4 Normal Forms

- A data model is said to exhibit a normalization state, called a ***normal form***.
- Three such states shall be discussed here: **first normal form (1NF)**, **second normal form (2NF)**, and **third normal form (3NF)**.
- One of the steps in refining a data model is to try to revise entities in first or second normal forms to third normal form.
- In moving from 1NF to 3NF, two types of dependencies must be eliminated:
 - ***Partial dependency***: Dependency based on part of a composite primary key
 - ***Transitive dependency***: Dependency among nonkey attributes

3.Normalization

3.4 Normal Forms – First Normal Form

First normal form has the following characteristics:

- 1. The key attributes are defined**
- 2. All attributes are dependent on the primary key(s)**
- 3. No composite attributes exist (i.e., each row/column intersection contains only one value, rather than a set of values)**

3.Normalization

3.4 Normal Forms – First Normal Form

Let's consider the following relationship describing the orders made by customers while identifying products, quantities ordered and prices :

PO_Report (Order_number, Order_date, Customer_number, Customer_country, Product_number, Product_price, Product_Quantity)

1- How to define The key attributes ??

Considering the functional dependency, ...

3. Normalization

3.4 Normal Forms – First Normal Form

PO_Report (Order number, Product number, Order_date, Customer_number, Customer_country, Product_price, Product_Quantity)

2. All attributes are dependent on the primary key(s) **ok**

3. No composite attributes exist (i.e., each row/column intersection contains only one value, rather than a set of values) **ok**

A table in first normal form can contain ***partial dependencies***

A table that contains partial dependencies will have data redundancy

This Data redundancy based on primary key attributes is eliminated when going from first normal form to second normal form.

3.Normalization

3.4 Normal Forms – Second Normal Form

The Table is said to be in second normal form 2NF if :

1. It is in 1NF

2. There are no partial dependencies based on the primary key(s)

PO_Report (Order_number, Product_number, Order_date, Customer_number, Customer_country, Product_price, Product_Quantity)

Let us check if a table is in a second normal form??

Product_Number → *Product_price*

Order_number → *Order_date*

Order_number → *Customer_number*

Order_number → *Customer_country*



The table is not in a 2NF

3.Normalization

3.4 Normal Forms – Second Normal Form

- A table is converted to second normal form by eliminating partial dependencies
- There are two steps in the process as follows:
 - 1. Write each key component that has partial dependency on a separate line and the original key on the last line**
 - 2. Define an entity name for each primary key (or composite key), and write the primary key attribute and dependent attributes of the entity**

3.Normalization

3.4 Normal Forms – Second Normal Form

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- There are two steps in the process as follows:
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 - 2. Define an entity name for each primary key (or composite key), and write the primary key attribute and dependent attributes of the entity**

PO_Report 1 (Order number, Order_date, Customer_number, Customer_country)

PO_Report 2 (Product number, Product_price)

PO_Report 3 (Order number, Product number, Product_Quantity)

3.Normalization

3.4 Normal Forms – Second Normal Form

- For this 2nd normal form, although there are no partial dependencies based on primary key attributes, there are dependencies within non-key attributes.

For example, considering the table :

PO_Report 1 (**Order_number**, Order_date, Customer_number, Customer_country)

Customer_number implies Customer_Country

- This dependency between non-key attributes is known as a ***transitive dependency***



The table PO_Report1 is not in a 3NF

3.Normalization

3.4 Normal Forms – Third Normal Form

When transitive dependencies are eliminated, a table has been converted to third normal form.

A table is in 3NF if :

- 1. It is in 2NF**
- 2. It contains no transitive dependencies**

3.Normalization

3.4 Normal Forms – Third Normal Form

- A table is converted to third normal form by eliminating transitive dependencies
- There are two steps in the process as follows:
 - 1. keep in the entity the attributes that depend directly on the key**
 - 2. Define in another table the attributes that are transitively dependent on the key. the transition attribute remains duplicated in the initial entity and becomes the primary key of the new entity**

3.Normalization

3.4 Normal Forms – Third Normal Form

PO_Report 1 (*Order_number*, *Order_date*, *Customer_number*, *Customer_country*)

1. keep in the entity the attributes that depend directly on the key

PO_Report 1-1 (*Order_number*, *Order_date*, *Customer_number*)

2. Define in another table the attributes that are transitively dependent on the key.
the transition attribute remains duplicated in the initial entity and becomes the primary key of the new entity

PO_Report 1-2 (*Customer_number*, *Customer_country*)

3.Normalization

3.4 Normal Forms – Third Normal Form

The conversion process results in the following table definitions:

Order (*Order_number*, *Order_date*, *Customer_number*)

Customer (*Customer_number*, *Customer_country*)

Product (*Product_number*, *Product_price*)

Product_Order (*Order_number*, *Product_number*, *Product_Quantity*)

4. Case Study in Data Modeling

We illustrate here the process of developing an E-R model using a hypothetical organization within the firm that is responsible for managing the materials inventoried by the firm. We will assume that the following information results from an interview process with key individuals in the organization:

1. The department is responsible for the inventory of three material types: **raw material**, **finished goods**, and **supplies**. *Each material, identified by a material ID, belongs to one material type. Each material type classifies many different materials.* These statements indicate the existence of two entities. One entity is the **MATERIAL_TYPE**, which defines a class of materials. The other entity is **MATERIAL**, with its primary key of material ID. At this point, we will refrain from including attributes in the diagram and focus on entities and relationships.

4. Case Study in Data Modeling

2. When a raw material or supply is ordered, it is done using a purchase order, where it appears on the purchase order detail (**PO_DETAIL**). Purchase orders are also used to order capital equipment and services. With our prior knowledge of the structure of a purchase order, we know that the material identification appears on the detail, not on the header of the purchase order. Therefore, MATERIAL is related to PO_DETAIL. Note that **MATERIAL** and **PO_DETAIL** have a **1 : M** relationship. The entity material is the unique set of all the different materials purchased and inventoried by the enterprise. Each material appears once in this set and is defined by the unique material ID. *Each of the materials may appear more than once on a PO_DETAIL*. MATERIAL is optional to the relationship because finished goods, one of the material types, is not ordered on a purchase order. Also, PO_DETAIL is optional in the relationship because some of the instances of PO_DETAIL are for capital goods and services, not material.

4. Case Study in Data Modeling

3. When the materials or supplies ordered on a purchase order arrive or when finished goods are produced, they are a tangible manifestation of the concept of MATERIAL. They are called ***material lots***, and each is assigned a unique identification called a *material lot number*. *For each material, there may be zero, one, or many material lots. Each material lot is the tangible manifestation of one and only one material.*

MATERIAL_LOT *has a relationship with MATERIAL and with PO_DETAIL.* Since it is possible that some instance of material may have been defined but never purchased or produced, the entity MATERIAL has an optional relationship to MATERIAL_LOT. However, *each material lot must participate in the relationship to material.*

The MATERIAL_LOT is the thing that is delivered to the enterprise to satisfy the order that appears on the PO_DETAIL. *Not all purchase order details are ordering material lots* since, as previously mentioned, capital equipment and services are also ordered on purchase orders. Some material lots are finished goods, so MATERIAL_LOT is optional in the relationship to PO_DETAIL.

4. Case Study in Data Modeling

4. The entity **MATERIAL_LOT** includes inventoried **raw materials**, **supplies**, and **finished goods**. These three categories are nonoverlapping and each instance of a material lot must appear in one and only one category.

This is an example of a **superclass-subclass hierarchy**. There is good reason to decompose MATERIAL_LOT into these subclasses. These subclasses do have some attributes in common.

For example, the attributes material ID, material description, quantity on hand, and so on, are common to all materials. However, there are clearly some attributes that are different. Purchased raw materials and supplies will be related to vendors and purchase orders on which they were purchased. This is not the case with finished goods. The finished goods are related to the **production lines** on which they were made, which is not relevant to raw material and supplies.

4. Case Study in Data Modeling

5. A **material lot** is stored in a **warehouse location**. *A warehouse location may store zero, one, or many material lots.*

If a material lot is too large to be stored in one warehouse location, it may have to occupy more than one location. A material lot may not be assigned to a warehouse location at some points in time.

Here we are interested in the material lots that occupy the locations in the warehouse. MATERIAL_LOT is optional in the relationship because it is not necessarily in warehouse storage.

Some warehouse locations may not be used for storage; they may be empty. Therefore, **WAREHOUSE_LOCATION** is optional with respect to this relationship. The cardinality of the relationship is **M : N**.

A material lot may be stored in more than one location, and a warehouse location may contain more than one material lot.

4. Case Study in Data Modeling

6. If a material lot is **finished goods**, it was produced on a **production line**. *A production line can produce more than one material lot of material, and each lot of finished goods will be produced on one and only one production line.*

This relationship is between a subclass of the entity **MATERIAL_LOT** and a new entity, **PRODUCTION_LINE**. This relationship is related to the subclass FINISHED_GOOD instead of the superclass MATERIAL_LOT.

When implemented in a table, the subclass FINISHED_GOODS will hold the foreign key *production line number*. A finished good lot is a physical manifestation of a product. It is possible that a product (the concept) may be produced on any one of several different production lines or assembly lines in the plant. However, when a finished good lot of that product is actually produced, it is produced on one, and only one, production or assembly line. Thus, a **1 : M** relationship exists.

4. Case Study in Data Modeling

7. A **production line** is made up of one or more pieces of **equipment**, each with its own *equipment ID*. Each piece of **equipment** *belongs to one and only one production line*. The relationship, which has cardinality 1 : M, is mandatory on both sides.

8. **Materials** may have **quality control tests** associated with them. This is true of some raw materials and all finished goods. Some raw materials and all supplies are not subject to any quality control tests. Some quality control tests are performed *on more than one material, and some materials require more than one quality control test*. Note that a quality control test is a description of a procedure used to perform the test. It is not the test itself. Actual tests are performed on material lots, the physical manifestation of material. An association between a description of a test procedure and the entity MATERIAL is considered. MATERIAL is optional to the relationship because some material is not associated with a test. On the other hand, **QUALITY_CONTROL_TEST** is mandatory in the relationship because it requires **MATERIAL** to which the test is associated.

To Conclude

- In this chapter we introduced the entity-relationship approach to developing a data model of the enterprise.
- The E-R modeling methodology as discussed in this chapter is the basis for other formats that are used for designing data models in industry.
- In industry, computer-aided software engineering (CASE) tools are widely used to document information system design projects and to help the designer to be more productive in executing those projects.